# AGGREGATE TRENDS PRESENTATION 4

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# Front Desk Analysis

Using CAR Data:

1. Identify front desk transfer paths in call flow
2. Analyze who returns and why

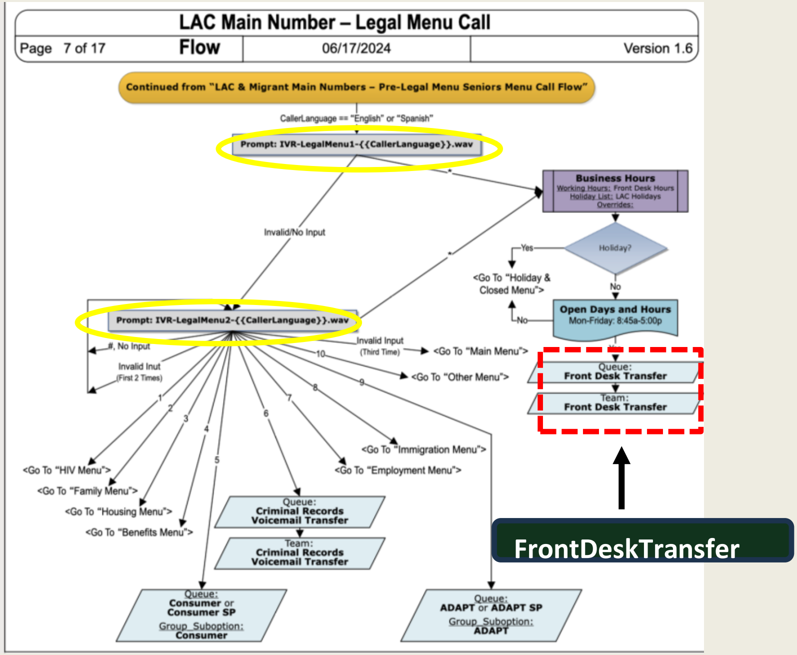
# Four District Front Desk Paths Drive 17,893 Calls, 85% Return to Main Menu

| Front Desk Type | Total Calls | % of FD Volume | % of All Calls | Return to Main Menu Rate |
|---|---|---|---|---|
| **FrontDeskTransfer** | 2,545 | 14% | 0.3% | 0% |
| **FrontDeskTransfer1** | 7,912 | 44% | 3.2% | 97.6% |
| **FrontDeskTransfer2** | 6,895 | 39% | 2.8% | 100% |
| **FrontDeskTransfer3** | 541 | 3% | 0.2% | 97.2% |

# FrontDeskTransfer is Embedded Within the Legal Menu (14% of Call Volume)

**1,968 callers (77%)** press star key on Legal Menu 1 to **request interpreter for languages** beyond English/Spanish; **445 callers (18%)** reach front desk via Legal Menu 2 routing.

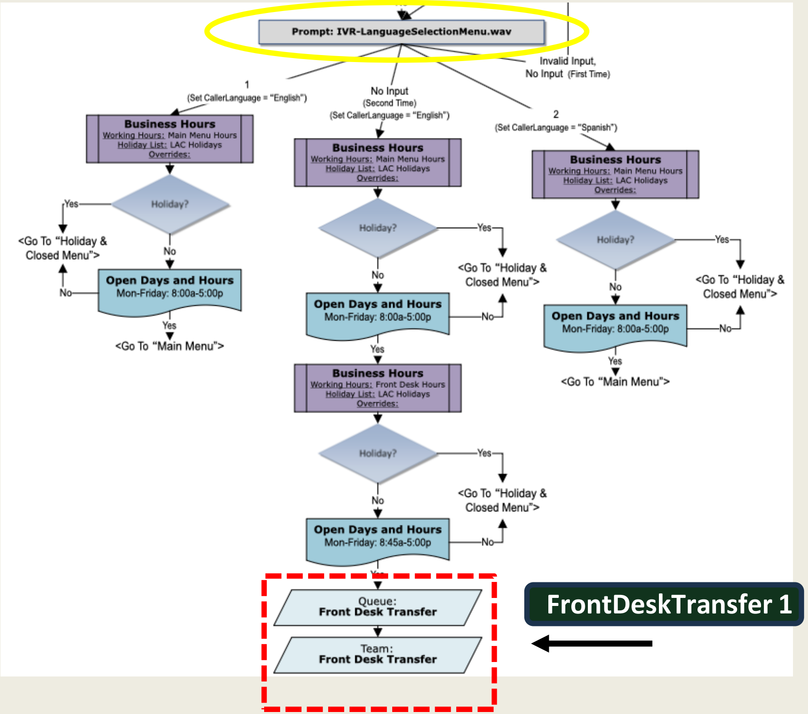
```
Total Sessions: 2,545
Analyzing all 2,545 sessions...

✅ Analyzed 2,413 sessions with valid 'before' data

📋 Most common activities BEFORE FrontDeskTransfer:
   1,968 ( 81.6%) - LegalMenu1
     445 ( 18.4%) - LegalMenu2

🏢 Most common EP Names BEFORE FrontDeskTransfer:
   2,543 (100.0%) - Legal Menu Telephony EP

📞 Most common Queue Names BEFORE FrontDeskTransfer:
  No queue data found

🔄 Main Menu Return Stats:
  Returned: 0 out of 2,545
  Return Rate: 0.0%
```



# FrontDeskTransfer1 Triggers at Language Selection (44% of Call Volume)

**7,407 callers (94%)** fail to select language after two prompts, **triggering automatic routing to front desk** with no context about caller needs or language preference.

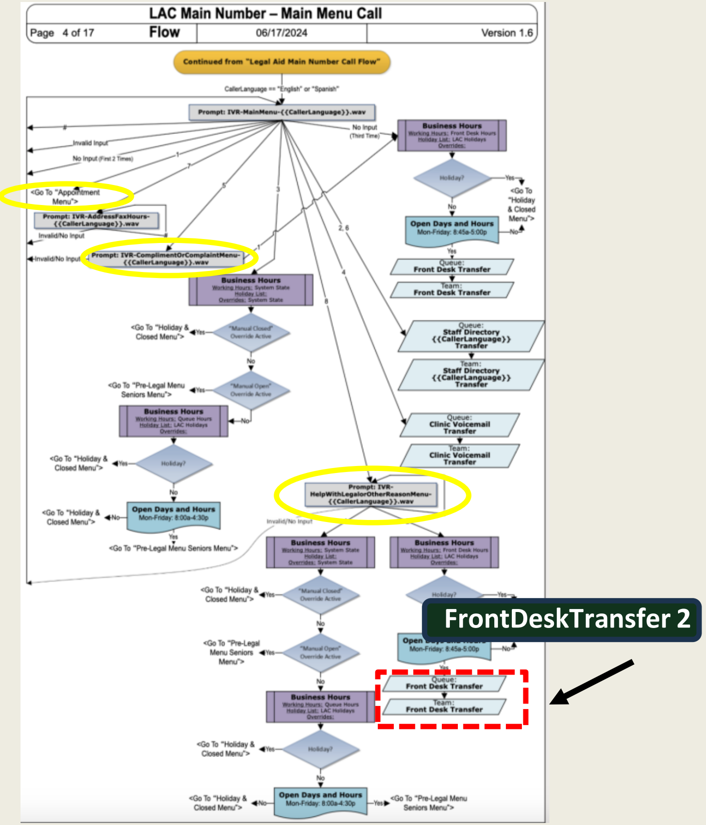
```
Total Sessions: 7,912
Analyzing all 7,912 sessions...

✅ Analyzed 7,407 sessions with valid 'before' data

📋 Most common activities BEFORE FrontDeskTransfer1:
   7,407 (100.0%) - LanguageSelectionMenu

🏢 Most common EP Names BEFORE FrontDeskTransfer1:
   7,912 (100.0%) - Main Number Telephony EP

📞 Most common Queue Names BEFORE FrontDeskTransfer1:
  No queue data found

🔄 Main Menu Return Stats:
  Returned: 7,725 out of 7,912
  Return Rate: 97.6%
```



# FrontDeskTransfer2 Routes From Main Menu Options (39% of Call Volume)

**6,895 callers (39% of front desk volume)** follow menu correctly by selecting **Appointment (41%), Legal Help (55%), or Complaint (4%) options**, but **all these routes lead to front desk.**

```
Total Sessions: 6,895
Analyzing all 6,895 sessions...

✅ Analyzed 6,542 sessions with valid 'before' data

📋 Most common activities BEFORE FrontDeskTransfer2:
  3,591 ( 54.9%) - HelpWithLegalorOtherReasonMenu
  2,663 ( 40.7%) - AppointmentMenu
    288 (  4.4%) - ComplimentOrComplaintMenu

🏢 Most common EP Names BEFORE FrontDeskTransfer2:
  6,895 (100.0%) - Main Number Telephony EP

📞 Most common Queue Names BEFORE FrontDeskTransfer2:
 No queue data found

🔄 Main Menu Return Stats:
 Returned: 6,895 out of 6,895
 Return Rate: 100.0%
```



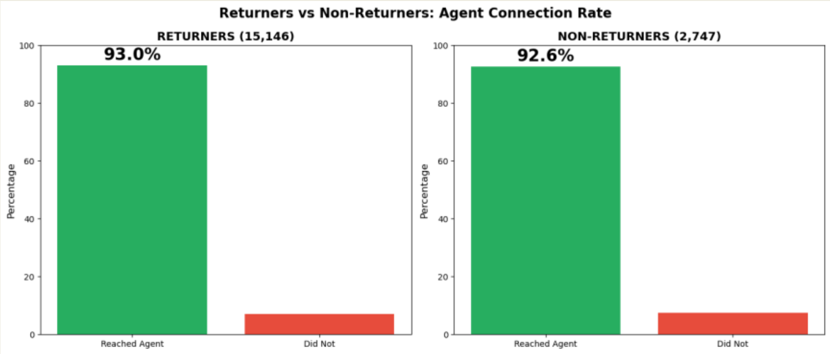
# FrontDeskTransfer3 Activates After Main Menu Timeout (3% of Call Volume)

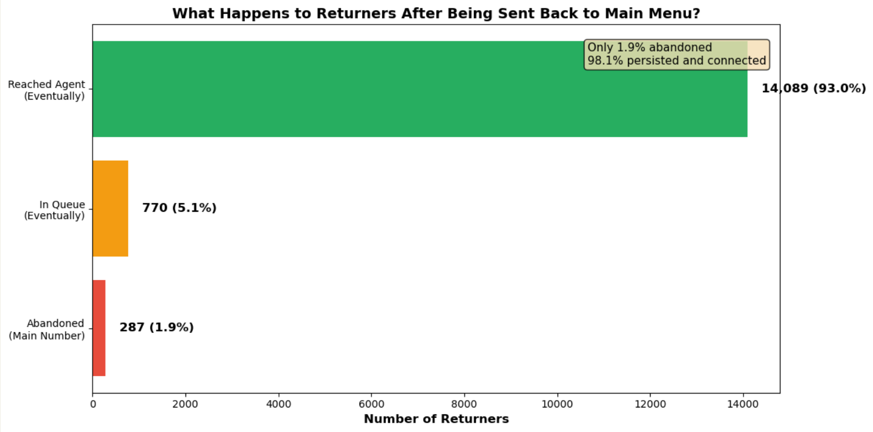
**510 callers (94%)** experience **Main Menu repeating 2-3 times** with 60-second intervals **before automatic routing to front desk.**

```
Total Sessions: 541
Analyzing all 541 sessions...

✅ Analyzed 510 sessions with valid 'before' data

📋 Most common activities BEFORE FrontDeskTransfer3:
    510 (100.0%) - MainMenu

🏢 Most common EP Names BEFORE FrontDeskTransfer3:
    541 (100.0%) - Main Number Telephony EP

📞 Most common Queue Names BEFORE FrontDeskTransfer3:
   No queue data found

🔄 Main Menu Return Stats:
   Returned: 526 out of 541
   Return Rate: 97.2%
```

```
============================================================
TEST 1: MainMenu Repetitions Before FD3
============================================================

MainMenu repetitions before FD3:
  Minimum: 2
  Maximum: 8
  Average: 2.9
  Median: 3

Distribution:
  2 repetitions:  127 sessions ( 23.5%)
  3 repetitions:  369 sessions ( 68.2%)
  4 repetitions:   32 sessions (  5.9%)
  5 repetitions:    6 sessions (  1.1%)
  6 repetitions:    5 sessions (  0.9%)
  7 repetitions:    1 sessions (  0.2%)
  8 repetitions:    1 sessions (  0.2%)
```

# 84.5% of Front Desk Callers Return to Main Menu But 93% of These Returners Eventually Reach Agents




- **14,089 out of 15,146 returners (93%) eventually connect with agents** after being sent back to main menu, nearly identical to non-returners' 92.6% success rate

- ”Return to main menu” represents **inefficient routing rather than failed calls**

- **Average Durations:**

  Returners: 4.39 minutes

  Non-Returners: 6.32 minutes

# Recommendation

1

## Update the language selection prompt and timeout handling.

**44% of all front desk calls (7,912 out of 17,893) come from the language selection timeout**, callers who don't press 1 or 2 get routed to the front desk after 2 minutes of waiting

- The menu says "*if you need an interpreter or relay service, please wait on the line"*
- Three Potential Fixes:

**(1)** Add *"press 0 if you need help"* option to minimize wait time
**(2)** Clarify if prompt is*: "select your language first, then wait for interpreter"* or if interpreters are for those who did not select language
**(3)** Analyze top third languages in call data and add direct option

# Front Desk Call Flow Performance Dashboard

1

**Volume Breakdown (Donut Chart)**

*Shows where the volume of FD calls is concentrated*

2

**Four Paths Comparison (Stacked Bar Chart)**

*FrontDeskTransfer, FD1, FD2, FD3*

3

**Trends Over Time (Line Chart)**

*X-axis: Month*
*Y-axis: Return rate %*

***Can also look at other time trends**

4

**Common Activities Before FD (Bar Chart)**

*Two side by side charts: Returners | Non-Returners & Shows: Language Selection Menu vs. LegalMenu 1, Main Menu vs. Seniors Menu*

5

**Common Activities After FD (TBD)**

6

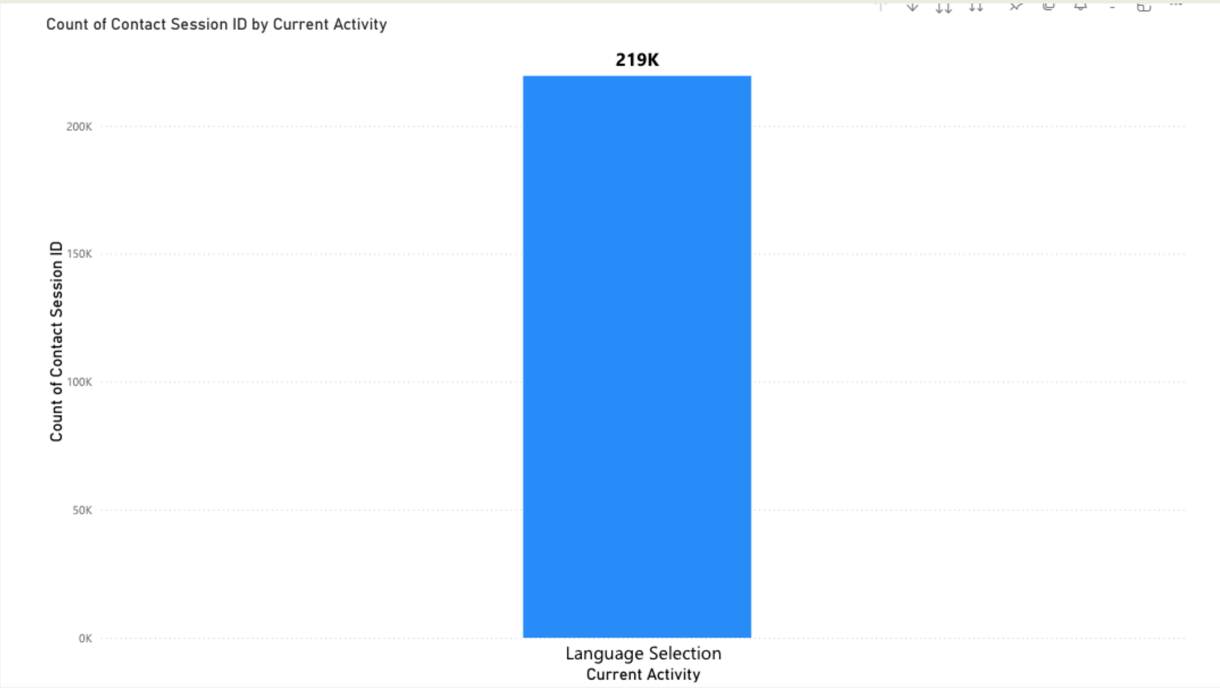
**Journey Length Comparison (Box Plot)**

*X-axis: Returners vs. Non-Returners*
*Y-axis: Average number of activities*

# Submenu Analysis

***Hierarchical Menu Usage Analysis: Power BI Drill-Down Dashboard***

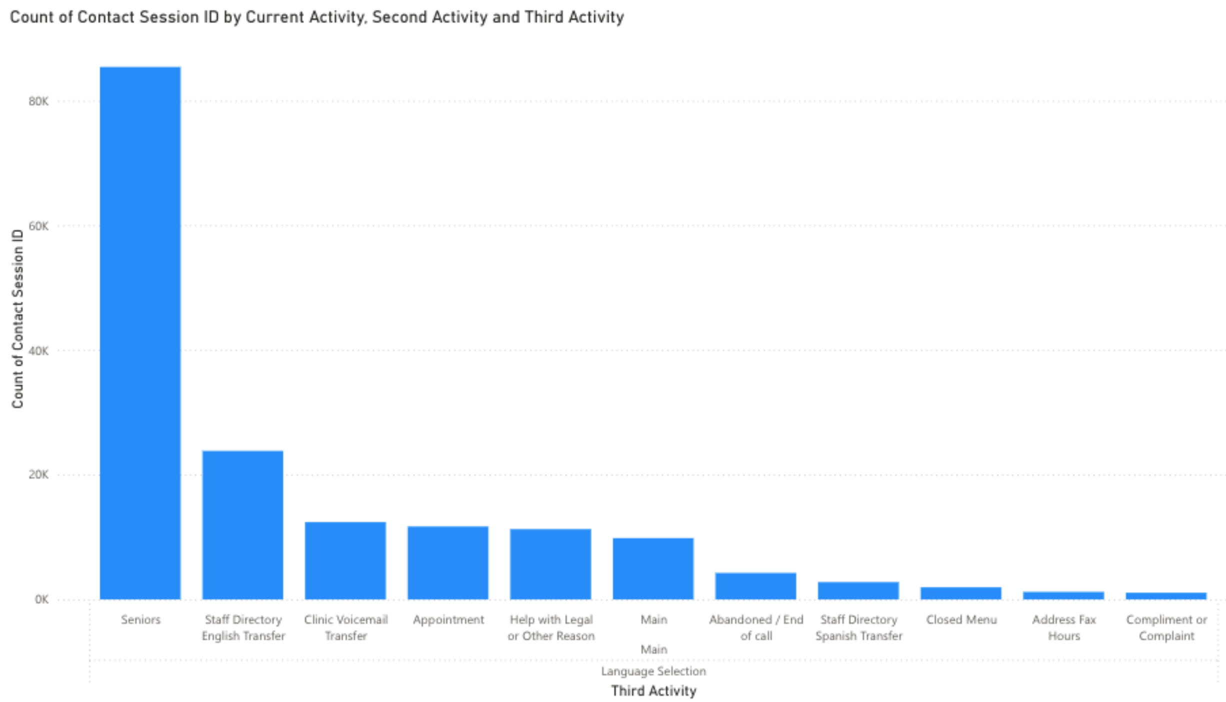
# Dashboard Updates from This Week




## What We Did:

- Combined activity variables in ‘Others’ category
- Added Activities 6-8 to make fully drilling down possible
- Separated Spanish options into their own menu categories
- Merged dashboards into one

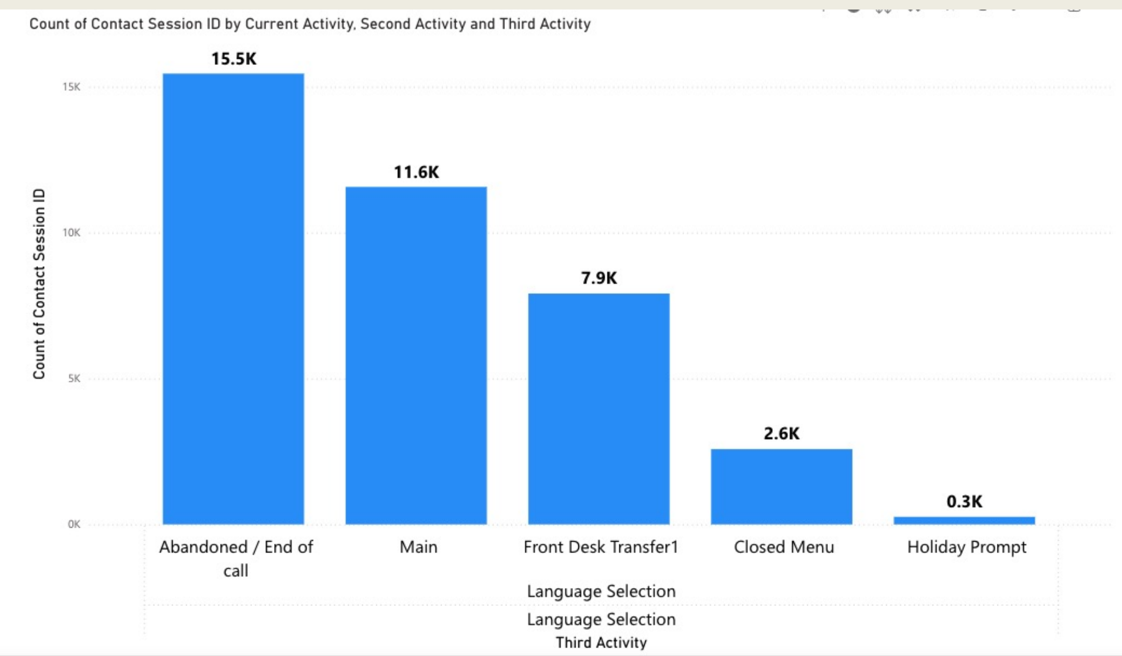
# Insight 1: Seniors Menu is a Major Routing Destination




## Insights:

- Over 50% of callers who reach the Main Menu go directly to the Seniors Menu
- Seniors Menu is one of the most frequently used paths → redesigning this could have a big impact
- Opportunity to streamline or prioritize support for senior callers

# Insight 2: Language Menu Confusion




## Insights:

- 41% of callers who listen to the Language Selection menu more than once end up abandoning the call
- Instructions to stay on the line if you need an interpreter may be unclear
- Callers may not get matched with an interpreter quickly
- Indicates a barrier to access for non-English or Spanish speakers

# Next Steps & Further Questions

**Next Steps**

- Update the Power BI dashboard to reflect the expanded activity hierarchy and visualize call paths across all menus
- Build KPIs to track statistics such as average activity depth per EP or call completion rates

**Further Questions**

1. Is it feasible that language-related barriers could contribute to call drop-off?

# Dashboard Design Ideas

***Laying out general structure of the final dashboard for interpretability and impact***

# Dashboard Goals

**INTERPRETABLE**

**USER-FRIENDLY**

- Understand what the data says regarding performance
- Understand the reasoning behind recommendations

# General Ideas

1. Organize in order of detail (high-level overview → performance details)
2. Focus on inefficiencies, insights, points of improvement, and recommendations
3. Incorporate text as needed to make the dashboard as quickly understandable as possible

# Structure Outline

1. **Performance overview + intro (1 page)**
    - mainly cards & text; high-level summaries like total % calls abandoned, total % calls resulting in call backs, total call count, etc.
        - perhaps a collection of cards that give 1-3 summary values (e.g., # of calls, % calls abandoned) for each of the top most popular menus (e.g., Family, Housing)?
    - blurb including info on dashboard contents, dashboard structure, and how to use the dashboard
2. **Call volume & traffic (>1 pages)**
    - blurb including notable insights + recommendations regarding call volume/traffic (e.g., call count, distribution of calls between different menus/times, queues)
    - contains visuals relating to call volume across menus & time ranges (e.g., date ranges, time of day)
3. **Call journeys (≥1 pages)**
    - visuals and blurbs about call duration, verbiage, points of confusion for customers, patterns in customer journeys
4. **Call outcomes (>1 pages)**
    - visuals and blurbs about abandonment rates, callback rates, voicemail rates, points of failure, etc.
5. **Additional considerations/details (≥1 pages)**
    - any additional visuals or insights that may be important but do not fully fit in any of the other categories

# Next Steps

1. Review all existing dashboards and corresponding insights
2. Begin compiling dashboards and revise organization as needed

# Appendix

**** Includes additional details and visuals from our analysis that were not included in the live presentation due to time constraints.***

# Appendix A

The following data analysis was completed in **Python** for **the All Calls Dataset**. Tatum's full code script is here.

```
In [3]:
# Create a function to identify which FrontDeskTransfer variant was used
def identify_front_desk_type(session_df):
    """Identify which FrontDeskTransfer (1, 2, or 3) was used"""
    fd_activities = session_df[session_df['Activity Name'].str.contains('FrontDeskTransfer', case=False, na=False

    if len(fd_activities) == 0:
        return None

    # Get the most common (or first) front desk transfer type
    fd_type = fd_activities['Activity Name'].iloc[0]
    return fd_type

# Mark sessions that went through front desk
sessions_with_fd = df_main[
    df_main['Activity Name'].str.contains('FrontDesk', case=False, na=False)
]['Contact Session ID'].unique()

print(f"Sessions with Front Desk interaction: {len(sessions_with_fd):,}")
print(f"Percentage of all sessions: {len(sessions_with_fd) / df_main['Contact Session ID'].nunique() * 100:.1f}%"

# Create a dataframe with one row per session that touched front desk
front_desk_sessions = []

for session_id in sessions_with_fd:
    session_df = df_main[df_main['Contact Session ID'] == session_id].copy()

    fd_type = identify_front_desk_type(session_df)

    front_desk_sessions.append({
        'Contact Session ID': session_id,
        'Front Desk Type': fd_type,
        'Total Activities': len(session_df),
        'First Activity': session_df.iloc[0]['Activity Name'],
        'Last Activity': session_df.iloc[-1]['Activity Name'],
        'Duration_Minutes': (session_df.iloc[-1]['Activity Start Timestamp'] -
                             session_df.iloc[0]['Activity Start Timestamp']).total_seconds() / 60
    })

df_fd_sessions = pd.DataFrame(front_desk_sessions)

print("\n" + "=" * 60)
print("FRONT DESK TYPE DISTRIBUTION")
print("=" * 60)
print(df_fd_sessions['Front Desk Type'].value_counts())
```